



M-Tech Production & Marketing

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Experiment 2

LAB TASKS

Internship Program — Batch 2026

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Date:	6 July 2026

1. Experiment 2 – Lab Tasks

File: *Experiment_2.py*

2. Viva Questions

1. What is type casting, and why is it necessary after using the `input()` function?

Type casting is the process of converting a value from one data type to another. It is necessary after using the `input()` function because `input()` always returns the entered value as a **string**. To perform mathematical operations such as addition, subtraction, multiplication, or division, the input must first be converted into a numeric data type using functions like `int()` for integers or `float()` for decimal values.

For example:

```
age = int(input("Enter your age: "))  
print(age + 1)
```

Without type casting, Python would treat the input as text and would generate an error during arithmetic operations. Therefore, type casting is essential whenever numerical calculations are required.

2. Differentiate between a List and a Tuple.

A **List** is a mutable data structure, meaning its elements can be modified, added, or removed after creation. In contrast, a **Tuple** is immutable, meaning its elements cannot be changed once the tuple has been created.

List	Tuple
Mutable	Immutable
Uses square brackets <code>[]</code>	Uses parentheses <code>()</code>
Supports adding, removing, and updating elements	Elements cannot be modified
Slightly slower	Faster and more memory efficient
Suitable for dynamic data	Suitable for fixed data

Example:

```
fruits = ["Apple", "Banana"]  
colors = ("Red", "Green", "Blue")
```

Lists are commonly used for shopping carts or student records, while tuples are suitable for storing fixed information such as days of the week or coordinates.

3. Explain the difference between a `for` loop and a `while` loop.

A **for loop** is used when the number of iterations is known in advance. It automatically iterates over a sequence or range of values.

```
for i in range(1, 11):  
    print(i)
```

A **while loop** is used when the number of iterations is unknown and execution continues until a specified condition becomes false.

```
num = 1  
while num != 0:  
    num = int(input("Enter a number: "))
```

The primary risk associated with a while loop is an **infinite loop**, which occurs if the loop condition never becomes false.

4. What is the purpose of the `return` statement in a function?

The `return` statement performs three important tasks:

- Returns a value to the function caller.
- Terminates the execution of the function immediately.
- Allows the returned value to be stored and used elsewhere in the program.

Example:

```
def add(a, b):  
    return a + b  
  
result = add(5, 3)  
print(result)
```

If a function does not contain a `return` statement, Python automatically returns **None**.

5. Why are dictionaries useful? Give a real-world example.

A **dictionary** stores information as **key-value pairs**, making data easier to organize and access using meaningful names instead of numeric indexes.

Example:

```
student = {  
    "name": "Ali",  
    "age": 20,
```

```
    "grade": "A"  
}
```

Data can be accessed using:

```
student["name"]
```

instead of numeric indexes.

Dictionaries are widely used for storing structured information such as student records, employee details, product information, and user profiles because they provide fast searching, updating, and retrieval of data.

6. Differentiate between = and == in Python.

The symbols = and == serve different purposes.

- **= (Assignment Operator):** Assigns a value to a variable.

```
name = "John"
```

- **== (Comparison Operator):** Compares two values and returns either **True** or **False**.

```
if name == "John":  
    print("Matched")
```

A common beginner mistake is using = instead of == inside conditional statements, which results in a syntax error.

3. Multiple Choice Questions (MCQs)

Q1. What is the index of the first character of a string?

Answer: (B) 0

Q2. Which loop is most suitable when the number of repetitions is already known?

Answer: (B) for

Q3. What is the output of len("hello")?

Answer: (B) 5

Q4. Which keyword sends a value back from a function?

Answer: (C) return

Q5. Which data structure stores information using key-value pairs?

Answer: (C) Dictionary

4. Intern Information

Field	Details
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